

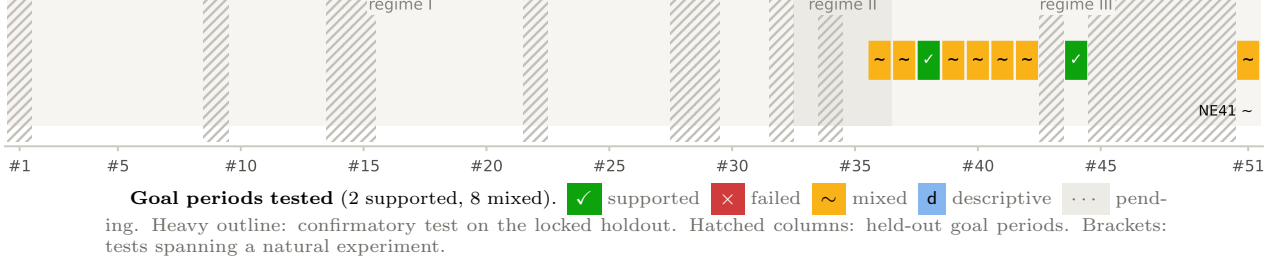
H44: Erasure makes agents busy but unproductive

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 62% Faithfulness 2/5 Usefulness 3/5

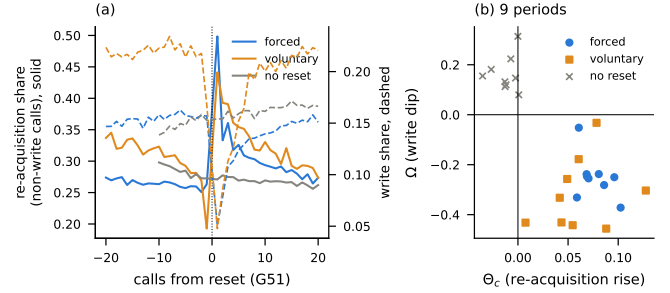
Bottom line. A forced context wipe causes a one-call re-reading spike with an ~ 8 -call tail ($\Theta_c = +0.106$ on blind-checked labels) and a -20% to -32% write dip; a 20-call cap would yield 12% less output per call than 40, and wiping does not pay as a loop breaker.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

The scaffold wipes an agent's context window at the 41-record cap and keeps its memory. Round 1 aligned 21,165 forced resets with no-reset pseudo-erasures over 1.24M calls in nine regime-III periods. The read share doubles at the first post-reset call. Among non-write calls it stays up for five calls ($\Theta_c = +0.079$, 9/9). Writes fall 26% and command loops break. There is no temperature pulse and no extra pull from the room. Round 2 tested four things. The call classifier holds on 300 blind labels, and Θ_c on the checked labels is $+0.106$. The sawtooth has two timescales and no linear ramp. A 20-call cap would lose 12% of output per call. Re-reading is working-set habit, and a wipe does not pay as a loop breaker.



Observables. Round 1. (a) G51: share of non-write calls that re-acquire (solid) and write share (dashed) against calls from a forced reset, a voluntary reset, or a no-reset boundary. (b) All nine periods: forced and voluntary resets sit at $\Theta_c > 0$, $\Omega < 0$; pseudo-erasures at $\Theta_c \leq 0$.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Model 02 as a kinetic multinomial spin. Each call has a category s_t (write, run, read, look, GUI, room read, talk, idle, ...). Its update has a baseline field h^0 , a context-held task field $h^{\text{ctx}}(k)$ and a persistence $J(k)$. Here k counts calls since the last reset; a wipe sets $k \rightarrow 1$, a quench of the task field. $R(k)$ is the re-acquisition share and $W(k)$ the write share at segment position k . $Y(L)$ is the output per call under a cap of L calls, with one summary call per consolidation. Model 04 asks which store restores output.

$$P(s_{t+1}=c) \propto e^{h_c^0 + h_c^{\text{ctx}}(k) + J(k)\mathbb{I}[c=s_t]}$$

$$y(k) = c + \beta(k-1) + A_1 e^{-(k-1)/\ell_1} + A_2 e^{-(k-1)/\ell_2}$$

$$Y(L) = \sum_{k \leq L} W(k) / (L+1), \quad \Omega = W_{+1..+10} / W_{\text{ref}} - 1$$

$$\Theta_c = \sum_{c'} w_{c'} [R_{\text{nw}}(\text{post} | c') - R_{\text{nw}}(\text{far} | c')]$$

H44-erasure-reacquisition-thrash · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H44-erasure-reacquisition-thrash/README.md · ratings: H44 agent (suggested)

Keep the context cap at 40 calls or more. In every well-sampled period output per call still rises at 40, and halving the cap to 20 loses about 12% (per call and per minute). Whether a longer cap would help is not identified. Do not use a forced consolidation to break loops. Most flagged loops are productive repetition, and a wipe costs a looping agent about 0.9 write calls over the next ten calls. After a wipe the agent re-reads the files it was just using, at its normal working-set rate. Re-reading is orderly, not thrash. Quote any cap cost with its reference window: the dip is -20% to -32% against pre-reset windows and -11% against the 40-call cycle mean. The best cap may depend on the model.

RESULTS VS PREDICTIONS

Prediction	Observed (pooled, 9 periods)	
P1 re-acquisition $\Theta_c > 0$	+0.079 [0.071, 0.086], 9/9	✓
P2 temperature pulse	$\Delta H -0.027$; $\Delta\sigma +0.018$	×
P3 write dip	$\Omega -26\%$ (8/9); commits -38%	✓
P4b susceptibility up	RR 0.97 [0.91, 1.04]	×
R1 classifier agrees; Θ_c checked	re-acq. κ 0.87; $\Theta_c +0.106$ [0.089, 0.122], 9/9	✓
R2 two timescales in $R(k)$ (G51)	$\Delta AIC +104$; spike 1 call, tail $\ell_2 = 8.2$	✓
R2 segment-long write ramp	G51 β n.s.; rise saturates ($\ell \approx 8.5$)	×
R2 no shorter cap helps	$Y(20)/Y(40) = 0.88$ [0.85, 0.91]; G51 edge 1.00	✓
R2 dip sign vs 5 references	-11% to -32% pooled; cycle mean $\sim$ n.s. in 4/9	
R3 erased files re-opened more	excess -0.039 $[-0.085, 0.007]$	×
R3 forced $>$ voluntary	excess $+0.040$ [0.017, 0.062]	✓
R4 wipe pays when looping	$E_{\text{loop}} -0.086$ $[-0.151, -0.022]$	×

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	2
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

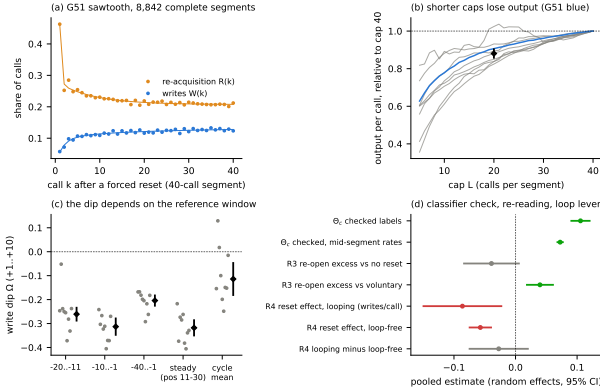
CAVEATS

The blind labels come from one rater (Claude) using tool names and arguments only. Clicks cannot be split into navigation and production. The two-timescale and ramp tests have power only in G51. Beyond 40 calls the optimum is not identified: regime I/II sessions longer than 40 calls end within about 50, so their late calls carry the end-of-session write burst. Re-open shares use file-path tokens from commands; the artifact-level check is n.s. The stuck-loop and gpt-5 results are post hoc. Agents choose re-acquisition paths. The reserved data are not run.

ROUND 2

Measure the cap effect directly at a scaffold change of the cap length, if one exists. Split looping events by failure state before and after reset. Run `confirm.py` after disclosure.

- **Essence.** how much of an agent's productive state lives in its context window rather than in memory or artifacts, and what an operator pays, and gains, each time the scaffold wipes it.
- **H44-R1** Validate the call classifier.
- **H44-R2** The sawtooth and the best cap.
- **H44-R3** What exactly is re-read.
- **H44-R4** Loop breaking as a lever.
- **H44-R5** Why no susceptibility rise.
- **H44-R6** Run `confirm.py` after committing the card and disclosing reuse.



Second figure. Round 2. (a) G51 complete 40-call sawtooths: re-acquisition and write share by call after a forced reset, with the two-timescale fit. (b) Output per call under a cap of L calls, relative to 40, for nine periods (diamond: pooled $Y(20)/Y(40)$). (c) The forced write dip against five reference windows (dots: periods; diamonds: pooled). (d) Pooled classifier check, recency-adjusted re-open excess and loop-lever effects.

GOAL PERIODS IN DETAIL

- G36 $\sim$ mixed (replication mixed (re-acquisition up, writes d...
- G37 $\sim$ mixed (re-acquisition up, write dip not significant, ...
- G38 ✓ supported (replication supported (re-acquisition up, ...
- G39 $\sim$ mixed (re-acquisition up, writes down, no entropy/swi...
- G40 $\sim$ mixed (re-acquisition up, writes down, no entropy/swi...
- G41 $\sim$ mixed (re-acquisition up, writes down, no entropy/swi...
- G42 $\sim$ mixed (re-acquisition up, writes down, no entropy/swi...
- G44 ✓ supported (re-acquisition up, writes down, switching ...
- G51 $\sim$ mixed (replication mixed (re-acquisition up, writes d...
- NE41 $\sim$ mixed (re-acquisition and output dip supported in 9/9...